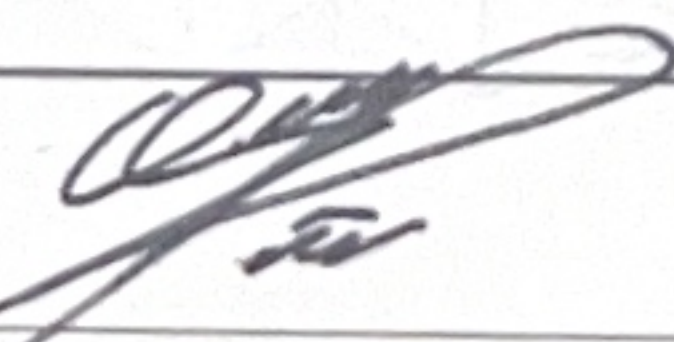
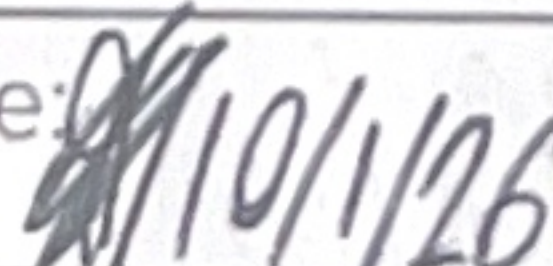
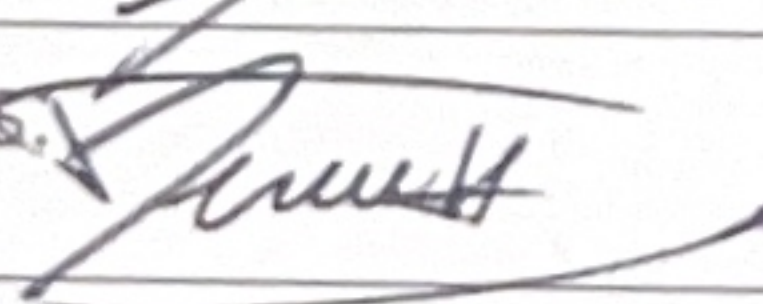


4. Three possible Directions

Idea and intended user/problem	Output or display	Manual user input	Automatic sensor	Actuator, mechanism, or hardware	Logic, processing, and control	Biggest risk or unknown
Idea #1: small automatic bridge Intended user: People managing small waterways or roads	LED's show whether the bridge is open or closed.	Push button to manually open/close the bridge or switch between automatic and manual mode	Proximity sensor detects an approaching object. Limit switches detect when the bridge is fully open or closed.	Servo motor raises/lowers bridge using a pulley system or hinge	Controller reads the distance sensor and limit switches, controls the servo, and changes the LED's based on the bridge's position	Making the bridge move smoothly and reliably without getting stuck or putting too much strain on the servo
Idea #2: An automatic night light Intended User: People who are scared of the dark or need a small light	LED's show on when it's dark and off when it's bright.	Push button to manually turn the light on/off or switch between automatic and manual mode.	Photocell sensor detects how bright or dark the environment is.	LED mounted to a small structure or enclosure made from available materials	Controller reads the light sensor and determines when the LED should turn on and off.	Getting the light sensor threshold set correctly so the LED doesn't constantly turn on and off.
Idea #3: An automatic door Intended user: People entering or leaving a building, especially when their hands are full.	LED's show whether the door is open or closed.	Push button to manually open/close the door or switch between automatic and manual mode.	Proximity sensor detects when someone approaches. Limit switches detect when door is fully open or closed	Servo motor opens and closes the door using a hinge and simple linkage. Door can be made with 3-D printer	Controller reads the proximity sensor, determines when to open or close the door, controls the servo, and uses the limit switches to confirm its position.	Making sure the door opens and closes smoothly and that the sensor does not cause it to repeatedly open and close

Signature: Date: 

Team Members:

Witness: 

Date: 10/1/26

N/A

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